

Sign Motion Generation by Motion Diffusion Model

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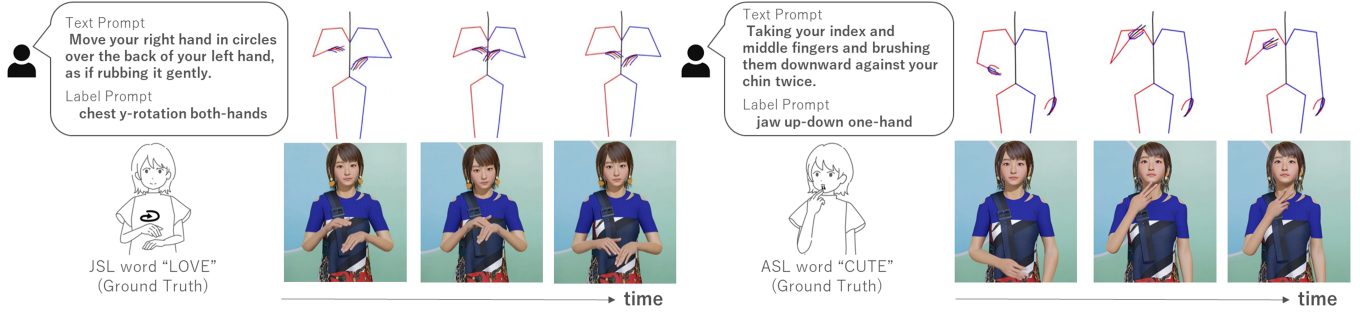


Figure 1: Generated sign motions from text and label prompt. The figure shows two samples using our method. Left/Right sides are generated sign motions for Japanese sign language and American sign language, respectively. For each sample, top left: text and label prompt, top right: output's motion, bottom left: the ground truth sign expression, and bottom right: the result of applying the motion to our avatar.

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1 ABSTRACT

We are researching ways to provide sufficient information to d/Deaf people, who use sign language(SL), through signing avatars [Uchida et al. 2023]. So far, we have developed a system that generates Japanese sign language(JSL) animations from Japanese news scripts. This system first translates input text into a sign sequence, and then generates animation by connecting each sign motion data pre-recorded by using motion capture. However, recording all sign motions used in SL incurs significant cost.

In the field of computer graphics, growing attention has recently been given to methods that generate human motion using diffusion

models [Tevet et al. 2023]. These methods enable "Text-to-Motion", generating motion data using text descriptions about human movement, without the need for motion capture. This could serve as a solution to the abovementioned high costs of recording motion data for every sign. To build these models, datasets such as KIT-ML [Plappert et al. 2016] and HumanML3D [Guo et al. 2022], containing 3D human motion and text, are primarily utilized. However, generating sign motion using these datasets is actually challenging because of a lack of hands' motion data and text explaining about hands' movements, which are crucial in SL.

Therefore, we are currently constructing a new dataset called the Sign Language-3D Motion Dataset (SLMotion3D) to enable sign motion to be generated from textual instructions. SLMotion3D is a collection of scripted sign motions, which, as of March, 2024, contains 2354 motions and 2398 texts/label descriptions (see Table 1). Motions of signs in JSL are recorded by optical motion capture. The motion data is represented as joints data including hand joints that follow the skeletal structure of our avatar. Texts and labels are verbal descriptions about sign motions extracted from the JSL dictionary [Yonekawa 2011]. Specifically, we annotated three elements as labels manually on the basis of the visual illustrations: the dominant hand's initial position, the dominant hand's initial movement direction, and the relationship between left and right hands.

In this study, we utilized SLMotion3D to train the Motion Diffusion Model (MDM) [Tevet et al. 2023] and performed sign motion

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Table 1: Comparisons of 3D human motion-language datasets.

Dataset	Hand's joints	Motions	Duration	Texts	Vocab	Labels
SLMotion3D	✓	2,354	1.10h	2,398	1,240	✓
KIT-ML [Plappert et al. 2016]	✗	3,911	10.33h	6,278	1,623	✗
HumanML3D [Guo et al. 2022]	✗	14,616	28.59h	44,970	5,371	✗

generation. To enhance the performance of the MDM during training, we used a two-step strategy: (1) pre-training using only motion data without text and label information and (2) incorporating label prompts in addition to text prompts during training. In our experiment, we found that our training method improves model performance. The details of our experiment results are in the supplementary file.

Figure 1 shows our results of sign motion generation from text and label prompts. The model trained using SLMotion3D enables body and hands to be controlled. As shown in the right side of Figure 1, the model can generate American SL signing for "CUTE", which is completely different from that in JSL. The trained model can be applied to not only JSL but also any SL as long as there are text instructions explaining how to move hands in English. Thus, we believe the advancement of this research could potentially extend SL avatar applications to multiple sign languages, increasing opportunities for information access for d/Deaf communities worldwide. Currently, the cases where the generated motions align

precisely with the target sign expression are still limited. Therefore, we will keep moving forward with our research to improve the performance of sign motion generation.

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